pdfencoding=unicodepdfpagemode=UseNone, bookmarksnumbered=truecolorlinks=true, urlcolor=blue, linkcolor=blue, citecolor=blue

Aalto University
School of Science

Master's Programme in Computer, Communication and Information Sciences

MPC in practice

MPC in practice

Tianxing Wu

Master's Thesis
2024

© 2024

This work is licensed under a Creative Commons "Attribution-NonCommercial-ShareAlike 4.0 International" license.

Aalto University
School of Science

Author Tianxing Wu

Title MPC in practice — MPC in practice

Degree programme Computer, Communication and Information Sciences

Major Algorithms

Supervisor Prof. Jukka Suomela

Advisor Hossein Vahidi

Date 10 July 2024 **Number of pages** 0 **Language** English

Abstract
Purpose:
The purpose of this thesis is to study how congested clique algorithms perform on modern supercomputers.
Methodology:
Studied how an MST algorithm performs on the Puhti supercomputer. The congested clique setting is simulated with MPI and CUDA.
Result:

Keywords For keywords choose, concepts that are, central to your, thesis

Preface

I want to thank Professor Jukka Suomela and my advisor Hossein Vahidi for their guidance.

I also want to thank my friends for keeping me sane and alive.

Otaniemi, 10 July 2024

Tianxing Wu

Contents

1 Introduction

TODO Why super computing, why it is important.
TODO What is the congested clique model.
TODO Relation between super computing and and the congested clique model.
TODO Why MST, what is representative in the algo
The algorithm is studied and implemented because it is representative and easy to implement. This algorithm is used because it is representative of the congested clique model.

- The computation graph is fully connected, In the algorithm, there is a step where messages are passed through every connection in the network.
- To comply with the bandwidth limit, the data from cluster leaders is spreaded to cluster menders before doing inter-cluster communication.
- The algorithm needs several iterations before its convergence.

In this work, The algorithm's compatibility is defined by firstly, how easy it is to translate the code:

- How to simulate local computation in a distributed setting.
- How to simulate message passing in the congested clique.
- How to deal with the bandwidth constraints in the congested clique.

Secondly, how the hardware is limiting the performance of the algorithm. Memory, CPU, GPU, network, which one is the bottleneck when the algorithm scales. How the algorithm performs compared to traditional sequential algorithms. By increasing the number of nodes, will it outperform the sequential algorithm with

- Faster speed.
- able to process more edges.

2 Literature review

2.1 Structure of the thesis

The thesis comprises the front matter, the main matter and possible appendices. The front matter in the required order is:

- a cover page,
- a page containing copyright information,
- the abstract page(s),
- an optional preface, and
- a table of contents.

If the thesis contains mathematical equations, give the list of symbols used to represent various quantities along with the mathematical operators used. The list must contain all the abbreviations used as well. Note that lists of figures and tables are not required.

The main matter, i.e. the actual thesis, begins after the list of symbols and abbreviations. Its structure follows the standard structure used in scientific writing. A typical division of the content is

1. Introduction (research objectives and questions)
2. Literature review (research made earlier in this area)
3. Research material and methods
4. Results / Findings
5. Discussion
6. Summary / Conclusions
7. References

The parts **??–??** in the list above are the text of your thesis. Often the discussion and summary (or conclusions) are combined in one chapter. The wording of the titles suggested above may differ. As a matter of fact, the titles above refer more to the type of content rather than being title suggestions. For example, the literature review of a thesis on electromagnetics could be under the title 'Electromagnetic theory and FDTD'. The background to the methodology used—FDTD here—could also be in this chapter. Note that the chapters and sections within the main matter are numbered, and that they appear in the table of contents. The references, or the bibliography, is also shown in the table of contents, but without a number labelling it.

The appendix or appendices, when necessary, are presented in the last part of the thesis. They contain things like questionnaires used in the study, [selected parts of]

data, derivations of mathematical results, a more detailed exposition of some aspect in the thesis, or code listings. Number them in the table of contents as follows:

A. The first appendix

B. The second appendix

Refer to a suitable guide on writing a thesis for a comprehensive description; there are plenty of them around. The instructions here are not exhaustive, and they concentrate more on how to use this template.

A bachelor's thesis is ideally about 20 pages long, excluding the appendices, whereas a master's thesis is about 60 pages long, again, not counting the pages in the appendices. The numbers pertaining to the length of your thesis given here are mere rules of thumb. Consult your school's thesis guidelines and your thesis advisor and/or supervisor for the appropriate length in your case.

2.2 Page numbering

Number the pages using Arabic numerals, placed in the centre of the page footer. The numbering begins from the cover page and runs continuously to the end of the thesis. Enter the total number of pages in the thesis—front matter, main matter and appendices, if present—in the appropriate field on the abstract page. Enter also the number of pages in the appendices, if present, in the same field on the abstract page as follows: total number of thesis pages / number of pages in the appendices.

Do not print the page numbers on the cover and the copyright pages; number all the other pages. Thus, the first page with a page number printed on it—the first abstract page—is '3'. The page numbering of appendices continues from that on the previous page.

2.3 Structuring the text in the main matter

2.3.1 Sections

You can structure your text by dividing it into sections with a suitable title. Well-devised sectioning clarifies your text, but beware of overusing them, since that will fragment and confuse your text. Do not use more than three levels of hierarchy in your sections. Notice that the section titles, just as the figure and table labels, are in a Helvetica clone, a sans serif, or grotesque, font. The main matter text is typeset in a Times clone, a serif font.

The sectioning macros available in LaTeX are

`\section{title}`,

`\subsection{subtitle}`, and

`\subsubsection{subsubtitle}`.

The formatting specifications are in appendix **??**. The fonts specified there are for Word documents, as specified by Aalto's visual guidelines. No corresponding fonts are officially specified for LaTeX. Hence, similar corresponding fonts provided by the `newtxtext` package are used in this template. Math fonts are those from `newtxmath`. The most important factor that influenced this choice of fonts was that the package outputs PDF/A-1b compliant pdf using `pdflatex`.

At least in technical texts, refraining from using the articles 'a' and 'the' as the first word in the title is common practice. Appendices are enumerated using capital letters. Begin every section on a new page. Begin a subsection on a new page only if the previous page is full. Similarly, begin every appendix on a new page.

2.3.2 Paragraphs

The paragraph that follows the section title is not indented, the paragraphs that follow are. Often, the body text is typeset as raggedright to avoid unnecessarily large white spaces between words. On the other hand, in many technical fields, the text is justified on both sides, but in such cases, hyphenation is used to maintain reasonable separation between words. This template uses justified text with automatic hyphenation. Write paragraphs so that they span more than three lines. Rethink your text, if you end up having one-line paragraphs; avoid them in your thesis—including the appendices. In addition, you may need to reconstruct your text if your paragraphs are too long. Avoid emphasising text using italics. Italicised text makes reading for dyslexic readers even more difficult.

When a number has a unit associated with it, ensure two things: the space between the number and the unit is smaller than the regular space (for example, 1 Hz) and the number and unit are not split over lines. You can simply use `\,` between the number and the unit without spaces between them to stop them from being split (`1\,Hz`) or use the package `siunitx`.

2.3.3 Equations

Equations are numbered using Arabic numerals, typically enclosed in parentheses, but not always. This template encloses equation numbers in parentheses. An example equation is

$$f(x) = a_0 + \sum_{n=1}^{\infty} \left(a_n \cos \frac{n\pi x}{L} + b_n \sin \frac{n\pi x}{L} \right). \tag{1}$$

Closely related equations like equations in a system of equations may be numbered as follows:

$$V_\mathrm{s} = R_\mathrm{s} i_\mathrm{s} + R(i_\mathrm{s} + i_\mathrm{b}) \tag{2a}$$

$$V_\mathrm{b} = R_\mathrm{b} i_\mathrm{b} + R(i_\mathrm{b} + i_\mathrm{s}). \tag{2b}$$

All equations need not be numbered. Remember punctuation after the equation; an equation is an integral part of your sentence and may need to be followed by punctuation marks. Many scientific journals do not use them after equations; then

Figure 1: A coil terminated in two LEDs with a magnet placed at its centre.

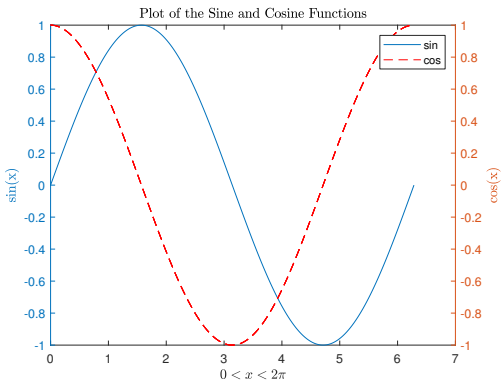

Figure 2: Example of a MATLAB graph.

again, many do. Punctuate the equations in your thesis. Numbering of equations in appendices consists of the appendix numbering alphabet followed by the equation number, which starts from one in each appendix. For example, the first equation in appendix A would be equation (A1) and the first two equations (A1)–(A2).

2.3.4 Figures and tables

Theses generally contain figures and tables, which have captions describing them. For figures, the convention is to place the caption after (below) the figure, as shown in figure **??** and figure **??**. Tables, on the other hand, have the caption above (before), as evident from table **??**. The label and number are typeset in the sans serif bold 12-pt typeface and the caption in the serif 12-pt normal typeface (see appendix **??**).

Table 1: An example table. Common animals classified into their classes.

Mammals	**Birds**	**Insects**
dog	crow	ladybird
cat	sparrow	ant
rat	tit	cockroach

Avoid placing figures and tables such that they are preceded or followed by just a line or two, because the reader can have difficulties in finding the line(s) of text and might even accidentally miss the text completely. Consider, for example, a page beginning with one line of text continued from the preceding page and that ends the paragraph. This is followed by a figure, then a short three-line paragraph, then yet another figure, followed by more text, which is positioned such that only two lines of text are on the page and the remainder is placed on the following page. The layout of this page is fragmented, and as a result, the reader might initially miss the top line of text. Consider using one of the following three layout possibilities: place the figures in the top part of the page followed by the text, or place the text in the top followed by the figures, or place all the text in the middle with one figure on the top and the other at the bottom. Base your choice on considering how easily the reader can follow your text (the content of the figures affects this), and whether the page as a whole looks ordered and pleasing to the eye.

Strive to place the referenced figure or table on the same page as where it is referenced. If this is difficult, place it on the following page, but not very much further. Using the options `b` (for bottom), `t` (for top), `h` (for here) and `p` (for separate page) judiciously in the `figure` and `table` environments, LaTeX does a pretty good job of placing figures and tables. However, placing figures (and tables) close to where they are referred to may be impossible when you have many pictures, so use common sense and keep the reader in mind when placing pictures and tables.

2.4 Referencing parts in the thesis

Cross-referencing in LaTeX is simple and straightforward: use `\label{marker}` to label the object with a number and then refer to the object using `\ref{marker}`. `marker` should be a unique string of characters. To help keep you organised with the various reference labels, be systematic when creating them for different objects in the thesis. For example, start labels for figures with `fig:`, for tables with `tab:`, `sec:` for sections, and `eq:` for equations, as done in this template. So, you could have, say, `\label{sec:ohmslaw}`, `\label{eq:ohmslaw}`, `\label{fig:ohmslaw}`, and `\label{tab:ohmslaw}`.

Perhaps you will have noticed that when referencing objects in the text, as is done for ‘figure’ earlier in the text, the label ‘figure’ in the cross-reference is not capitalised. Often, you see the capitalised label ‘Figure’. In such cases, the label together with the number, for example ‘Figure 2’, is considered a proper noun, hence the capitalisation. Although this template uses the lowercase form, you can choose how you write cross-referencing labels: ‘figure’ or ‘Figure’—you will do well to consult your advisor and supervisor about this; they may have strong opinions on the matter. However you decide to write cross-referencing labels—figure, table, section or equation—be consistent.

Do not abbreviate any of the labels, for example ‘figure’ to ‘fig.’ or ‘Fig.’. Journals do this because the amount of available space for an article is limited; space is not an issue in your thesis. You can refer to more than one figure, for example, by saying figures **??** and **??**. Use endash ‘`--`’ to specify a range: sections **??**–**??**.

To reference a section, place the section label directly after the sectioning command of the section to be referenced. For example:

```
\subsection{Ohm's law}
\label{sec:ohmslaw}
...
...as discussed in section~\ref{sec:ohmslaw}, the resistance...
```

The tilde placed between 'section' and '`\ref{sec:ohmslaw}`' not only puts a normal space between the two, but also binds them so that they are not split over separate lines if the two occur at the end of a line. Put a tilde between all labels and the reference number.

For figures and tables, place the label text after `\caption`. For simple equations, the label is placed after `\begin{equation}`. You can reference individual items such as equation (**??**) in a grouped equation enviroment like `subequations` or an item in an itemised list like item **??** on page **??** or the system of equations (**??**) by putting a `\label` appropriately.

2.5 Citation

The list of reference must be carefully made, following one of many different styles. Which you choose depends on the field of your thesis. However, citing using footnotes is not allowed in your thesis[1]. A short reference list can have entries in the order in which they are cited, but if your list is long, arrange the entries in alphabetical order. Read appendix **??** for a fairly thorough discussion of how to cite work in academic writing and make a bibliography or list of references. Remember that referencing is a very important part of your thesis, so be meticulous and thorough.

2.5.1 Bibliography or list of references

Find here a list of some of the typical entries in the bibliography and the information they must contain. For a more comprehensive list, see [**?**]. See appendix **??** for details on how this information must be typeset in different reference styles.

A book entry must contain the following:

- author(s)
- title of the book
- edition, if there are more than one
- publisher
- place of publication
- time of paublication (typically year)
- name of series, if the book is part of a series

[1] Adding comments in footnotes is not allowed either.

References [**?**]–[**?**] are examples of books in the bibliography. Refering to a particular place in a book, for example like this, [**?**, s. 83–124], is a good practice. The reader can thus find the referred matter easily.

The following must be found in an article entry:

- author(s)
- title of the article
- name of the journal or magazine where the article is published
- volume number of the journal or magazine
- issue number of the journal or magazine
- year of publication
- page numbers of the published article

References [**?**]–[**?**] are examples of articles in a bibliography.

An edited book or compilation of contributions, typically chapters or articles, from several authors having one or more editors or a compiled conference proceedings must have the following information:

- author(s) of the chapter or part
- title of the chapter or part
- mention the kind of publication 'In book', or for conference proceedings with no editor information use 'In' before the name of the conference proceedings
- editor(s) of the compilation identified with (Ed.) or (Eds.)
- name of the compiled book or conference
- for a conference article, the time and place where the conference was held
- edition, if more than one
- place of publication
- publisher (may not exist for conference proceedings)
- time (year) of publication
- pages of cited contribution
- name of series, if the book is part of one

References [**?**]–[**?**] are examples of references to a part of a compilation.

Theses must have the following information:

- author
- title of the thesis
- type of thesis (bachelor's, master's doctoral, other)
- name of university or institution
- name of the department, faculty or degree programme
- location/place of the university or institution

- year of publication

References [**?**]–[**?**] are examples of theses in a bibliography.

Standards must have the following:

- standard identifier and number
- name of the standard
- edition, if not the first
- place of publication
- publisher
- year of publication
- number of pages

Reference [**?**] is an example of how a standard is entered in a bibliography.

An interview entry must have the following details:

- name of the interviewee
- position or t academic and/or professional title
- organisation where the interviewee is placed
- address of the organisation
- a mention that the entry is an interview along with the date of the interview

Reference [**?**] is an example of an interview entry in a bibliography.

Most scientific articles are available online nowadays and many also in printed form. Articles from journals, magazines or other publications available only online must have the following details in the bibliography:

- author(s)
- name of the article
- name of the publication (journal or magazine)
- type of publication
- edition or volume
- volume number and issue details of publication and article
- year of publication along with details on possible updates: Updated, date of update
- mention 'Accessed' and date when accessed
- mention 'Available' and URL or mention 'DOI' ja DOI number (DOI=Digital Object Identifier).

References [**?**]–[**?**] are examples of how theses available online are presented in a reference list. Additionally, references [**?**] and [**?**] available both online as well as in print, so the details are presented as for the corresponding entry in print and the details of the online version are also given. Reference [**?**] is available only online, so

only those relevant required details are given. Unfortunately, the details on the edition, publication name and number are often unavailable and so cannot be presented in the reference list.

For a thesis that is only available online, give the following details:

- author
- thesis title
- type of medium (online, DVD, CD-ROM etc.)
- type of thesis (bachelor's, master's doctoral, other)
- name of institution or university
- name of the department, faculty or degree programme
- location/place of the university or institution
- year of publication
- date of citation
- availability: mention 'Available' and give the URL and/or mention 'DOI' and the DOI number

Reference [**?**] is an example of how a thesis available in electronic form is entered in a bibliography.

Reference [**?**] is an example of an online stand-alone article, one not associated with any journal or magazine. Such an article is considered a work of its own. The details required of such a stand-alone webpage are:

- author(s)
- title
- mention 'Updated' and date of update
- mention 'Accessed' and date of access
- mention 'Available' and the URL

If the article spans more than one page, the reference list must contain the URL of the page pointing to the entire article, typically its homepage, unless the intention is to cite a particular page in the article.

3 Research material and methods

This part is the core of your work, where you explain the methodological choices you made, its limitations, how you pick your research material or subjects, the implementation of your study and the methods used. This section determines the methodological strengths and weaknesses of your thesis. Any earlier description of the method should limit itself to work done earlier by others. Here you tell your reader what you have done.

4 Results

Present the results of your study here and answer the research questions, asked earlier in the thesis (in the introduction, perhaps), this study strives to answer. The scientific value of your work is measured by the results you obtain along with the arguments you give to back the answers to your research questions.

Be critical of the significance of your results. You may critically scrutinise the results and your interpretation of the results here, or you may do so later in the chapter with the discussion of your work or in the conclusions part.

This part should discuss how reliable the data used in the study are. You may discuss the reliability of the conclusions drawn from the study either in this chapter or later in the discussions part. You may have the discussion in a chapter of its own, separate from the summary or conclusions.

5 Summary/Conclusions

This is where you tie up any loose ends. Tell your reader briefly and clearly what you have done, what you have discovered, and the value of your discovery in the context of similar work done earlier. Draw clear conclusions regarding the research problem, sub-problems or hypotheses. You also discuss future lines of study and new questions your study might have posed.

As the author of the thesis, you alone are responsible for ensuring that the layout, form and structure of your thesis adheres to the guidelines outlined by your school. This template aims to help you meet these requirements.

References

[1] Zvi Lotker, Boaz Patt-Shamir, Elan Pavlov, and David Peleg," *SIAM Journal on Computing*, 35(1):120–131, 2005. doi.org/10.1137/S0097539704441848